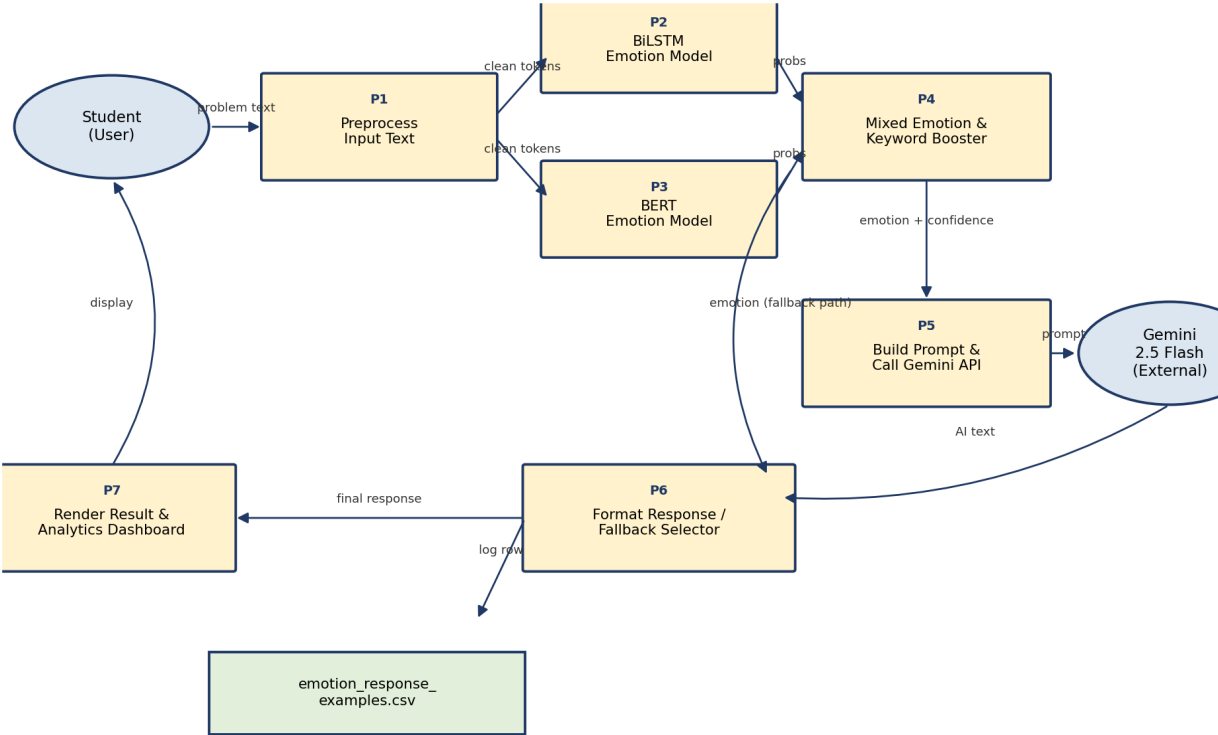


Data Flow Diagram

Date	15 July 2026
Team / Submission ID	
Project Name	Emotion Detection & Learning Support Engine
Submitted By	Venu Gopal (Sole Team Member / Team Lead)
Maximum Marks	2 Marks

Data Flow Diagram (Level 1)

The diagram traces how a student's free-text problem description moves through preprocessing, the two emotion classifiers, the Gemini generative layer, and back to the student, with a CSV data store used for logging.



DFD Symbol Legend

Symbol	Name	Description
Oval	External Entity	Student (sends the problem text, receives the response) and the Gemini API (external generative AI service).
Rounded rectangle (numbered)	Process	P1 Preprocess Text, P2 BiLSTM Model, P3 BERT Model, P4 Mixed Emotion & Keyword Booster, P5 Build Prompt & Call Gemini, P6 Format Response, P7 Render Result & Dashboard.
Solid rectangle	Data Store	emotion_response_examples.csv, which persists every logged interaction (field, problem, emotion, confidence, response, timestamp).
Labelled arrow	Data Flow	Movement of data such as raw text, cleaned tokens, probability scores, prompts, and the final AI/fallback response.